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Batch 15

Algorithm

1. Start.
2. Read the size `n` of the array.
3. Declare the variable-length integer array `arr[n]`.
4. Read `n` elements into the array using pointers.
5. Pass the address of `arr` and `n` to `previous_greater()`.
6. For each element `arr[i]`:
   - Set `found = -1`.
   - Start searching from index `i - 1` toward index `0`.
   - If `arr[j] > arr[i]`, store `arr[j]` in `found` and stop searching.
   - Print `found`.
7. Separate each result with a comma.
8. Stop.

Test Case Table

Test	Input Array	Expected Output
1	`4 5 2 25`	`-1, -1, 5, -1`
2	`13 7 6 12`	`-1, 13, 7, 13`
3	`1 3 2 4`	`-1, -1, 3, -1`
4	`1 2 3 4 5`	`-1, -1, -1, -1, -1`
5	`5 4 3 2 1`	`-1, 5, 4, 3, 2`
6	`10`	`-1`
7	`5 5 5 5`	`-1, -1, -1, -1`

C program

```
#include <stdio.h>
```

```
void previous_greater(int *arr, int n)
{
    int i, j, found;

    for (i = 0; i < n; i++)
    {
        found = -1;

        // Search from the nearest element on the left
        for (j = i - 1; j >= 0; j--)
        {
            if (*(arr + j) > *(arr + i))
            {
                found = *(arr + j);
                break;
            }
        }

        printf("%d", found);

        if (i < n - 1)
            printf(", ");
    }

    printf("\n");
}

int main()
{
    int n, i;

    printf("Enter the size of array: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d elements:\n", n);

    for (i = 0; i < n; i++)
    {
        scanf("%d", arr + i);
    }

    printf("Previous greater elements:\n");
    previous_greater(arr, n);

    return 0;
}
```

ExecutionThe search is:

- 13 → nothing on its left → −1
- 7 → nearest greater on left is 13
- 6 → check 7; 7 > 6 → 7
- 12 → check 6, then 7, then 13; 13 > 12 → 13

```
 Last login: Mon Oct 19 00:14:40 on ttys000
[bhavya@Bhavyas-MacBook-Air ~ % nano day52.c
[bhavya@Bhavyas-MacBook-Air ~ % clang day52.c -o day52
[bhavya@Bhavyas-MacBook-Air ~ % ./day52
 Enter the size of array: 4
 Enter 4 elements:
 13 7 6 12
 Previous greater elements:
 -1, 13, 7, 13
[bhavya@Bhavyas-MacBook-Air ~ % ./day52
```